

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

Q.23 Write short notes on:

- a) Concept of jigs and fixture in batch and mass production.
- b) Explain GO and NOT GO Gauges

Q.24 Define press tool, explain in details the elements of press tools with diagram.

Q.25 What do you understand by integrated operations of press tool? Also classify the integrated operations.

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Roll No.

2nd Year / Advance Diploma in Tool and Die Making

Subject : Tool Design-I (Press Tool, Jigs and Fixtures)

Time : 3 Hrs.

M.M. : 60

SECTION-A

Note: Multiple choice questions. All questions are compulsory (6x1=6)

Q.1 Which dies is used for more than one cutting operation per stroke?

- a) Blanking Die b) Piercing Die
- c) Compound Die d) Progressive Die

Q.2 Punching the number of holes in a sheet is known as

- a) Performing b) Drawing
- c) Half of the die d) Notching

Q.3 What is the effect of poor lubrication in dies?

- a) Reduced wear
- b) Increased friction and heat
- c) Improved accuracy
- d) Lower maintenance cost

- Q.4 A fixture does not
- a) Hold the workpiece
 - b) Locate the workpiece
 - c) Guide the tool
 - d) All of them

- Q.5 Name of the material used for the die
- a) Grey cast iron b) Copper
 - c) Aluminium d) Carbide

- Q.6 In auto CAD the 3-D commands on the modelling toolbar include.
- a) Sphere b) Extrude
 - c) Box d) All of them

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (6x1=6)

- Q.7 Write two applications of combination die.
- Q.8 Define Shearing.
- Q.9 Describe mass production.
- Q.10 Name any two specifications of press.
- Q.11 What is die cushion?

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- Q.12 _____ is the component in press tools used to prevent the work piece from sticking to the punch.

SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

- Q.13 What do you understand by B.O.M. Explain it with an example.
- Q.14 Define CAM for manufacturing and development of press tool.
- Q.15 Write the work sheet principle of sheet material.
- Q.16 Explain the geometrical relationship between component and punch.
- Q.17 Explain the function of different elements of drawing operation.
- Q.18 Explain estimation method in tool design with its two application.
- Q.19 Explain various material handling equipment.
- Q.20 Describe the Taylor's principle of designing limit gauge.
- Q.21 Explain the concept of safety and maintenance for a die.
- Q.22 Explain various material handling equipment.

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